Map Compressor

Turn large drone maps into small, see-through overlays, right inside ATAK

ATAK Plugin User Guide · Version 0.1.1 · November 2026

What it does

Drone mapping software (PIX4Dreact, PIX4Dengine, DroneDeploy, WebODM and others) produces orthomosaic maps that are often **hundreds of megabytes**. Those files are slow to send, too big for many phones and tablets, and the empty area around the flight shows up as a **black box** in TAK.

The Map Compressor plugin shrinks the map on your Android phone or tablet, inside ATAK, and adds it to the map. It keeps the map in the right place, keeps the empty border **see-through** so only your imagery shows, and makes a file small enough to share with your team.

223 MB → ~31 MB

A real fire-scene drone map, converted on a phone in about a minute

See-through

No black box around the imagery. The basemap shows around it.

100% on-device

No account, no cloud, no upload. Safe for sensitive imagery.

What you need

📱 ATAK-CIV 5.6, 5.7 or 5.8

On an Android phone or tablet. The normal Play Store or TAK.gov version is fine.

🗺️ A drone map

A GeoTIFF file (`.tif`) exported from your drone mapping software.

📡 Nothing else

No account, no internet and no other apps. It uses ATAK's own map tools.

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Download

Click the link for your ATAK version, ideally on the phone or tablet itself. The download starts straight away; no account or sign-up is needed.

📱 **ATAK 5.8**	Download Map Compressor for ATAK 5.8	`github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/ATAK-Plugin-MapCompressor-0.1.1--5.8.0-civ-release.apk`
📱 **ATAK 5.7**	Download Map Compressor for ATAK 5.7	`github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/ATAK-Plugin-MapCompressor-0.1.1--5.7.0-civ-release.apk`
📱 **ATAK 5.6**	Download Map Compressor for ATAK 5.6	`github.com/AM-Ripple2/map-compressor/releases/download/v0.1.1/ATAK-Plugin-MapCompressor-0.1.1--5.6.0-civ-release.apk`

Which ATAK download do I need?

Open ATAK's **About** screen and look at the version number. Only the first two numbers matter:

ATAK shows...	Download
5.8.x.x	ATAK 5.8
5.7.x.x	ATAK 5.7
5.6.x.x (for example 5.6.0.13)	ATAK 5.6

✅ **Signed by TAK.gov**

All three ATAK downloads went through the TAK.gov third-party plugin pipeline. They load in the normal ATAK-CIV from the Play Store or TAK.gov. You don't need a developer version of ATAK.

Downloading on a computer instead of the device?

Save the file, then copy it to the phone or tablet with a USB cable, email, or a shared drive. Put it somewhere easy to find, such as the **Download** folder.

Newer versions

To check for a newer version later, open github.com/AM-Ripple2/map-compressor/releases/latest and click a file name under **Assets**.

📱 Using the ATAK plugin

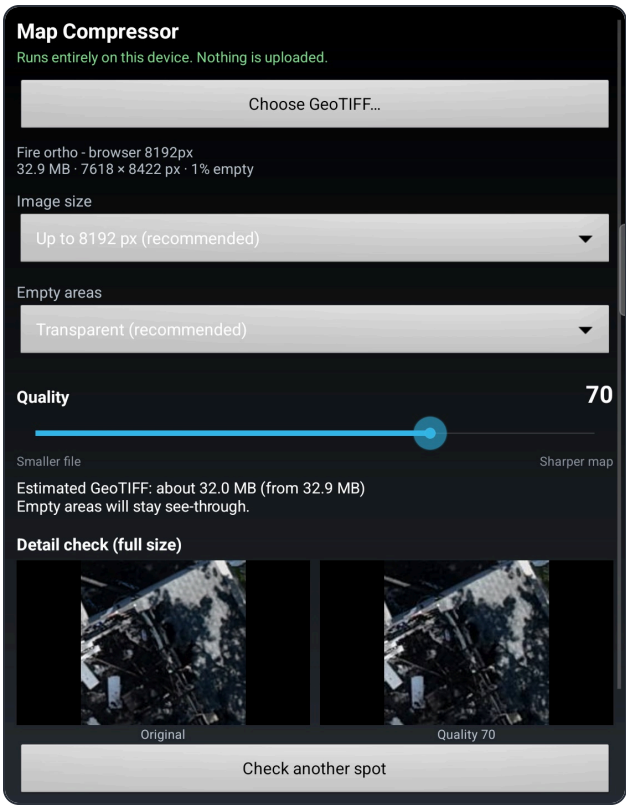
Install (once)

1. Download the file for your ATAK version (see page 2) on the phone or tablet.
2. Tap the downloaded file, then tap **Install**. If Android asks, allow installs from your browser or file manager.
3. Open ATAK and go to **Settings** → **Tool Preferences** → **Plugins**. Find **Map Compressor** and load it.
4. ATAK may show that the plugin was signed through the TAK third-party pipeline. That's expected; accept it.

Updating later: install the new file over the old one. You don't need to uninstall first.

Convert a drone map

1. Put the drone map file (`.tif` or `.tiff`) on the device, for example in the **Download** folder.
2. In ATAK, open **Map Compressor** from the toolbar or the **Tools** menu. A panel opens on the side (right).
3. Tap **Choose GeoTIFF...** and pick your file. Below the button, the panel shows the map's name, file size, pixel size and how much of it is empty.
4. Leave the settings on the recommended defaults, or adjust them (see page 5):
   - **Image size:** Up to 8192 px (recommended)
   - **Empty areas:** Transparent (recommended)
   - **Quality:** 70. Drag toward *Smaller file* or *Sharper map*. The **Estimated GeoTIFF** size updates as you drag.
5. *Optional:* the **Detail check** shows the same spot from the *Original* and the compressed version side by side, at full size. Tap **Check another spot** to look somewhere else.



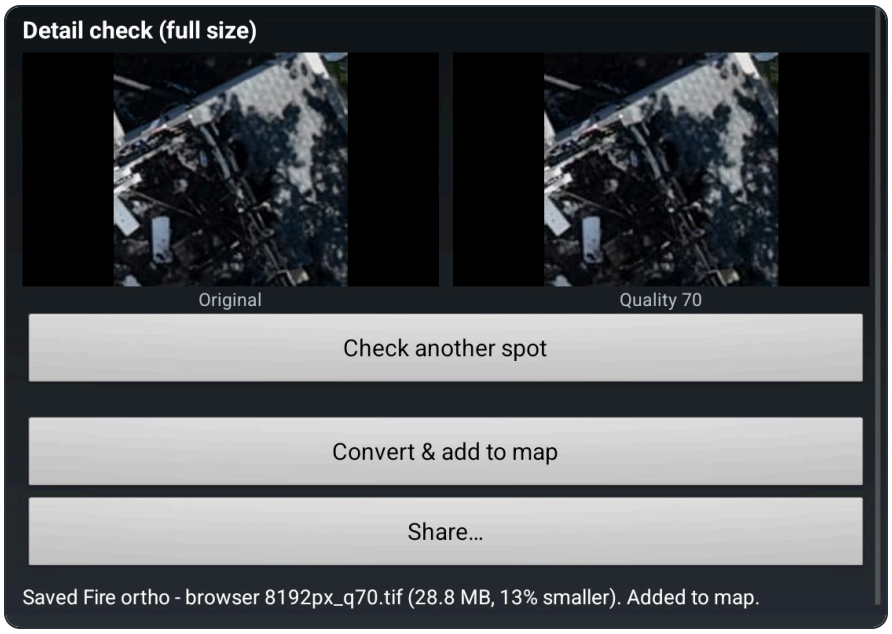
The Map Compressor panel with a map loaded

📱 Using the ATAK plugin (continued)

Convert a drone map

6. Scroll to the bottom of the panel and tap **Convert & add to map**. A progress bar shows how it's going; a large drone map takes about a minute.

7. When it's done, the overlay is added to the map and the map zooms to it. The last line of the panel shows the saved file name, its size and how much smaller it is.



Bottom of the panel after converting. This example input was already compressed, so it only shrank 13%; the original 223 MB drone map shrank by 86%.

Where is the file saved?

In the device's `atak/grg` folder, named after the map and the quality, for example `FireScene_q70.tif`. ATAK lists it with your other overlays.

Share it with your team

Tap **Share...** to send the map. See page 6 for the options.

Settings explained

Image size

Limits how many pixels wide or tall the finished map can be. Smaller means a smaller file.

Option	When to use it
Up to 8192 px (recommended)	The best balance for most drone maps: sharp at normal zoom, and a much smaller file.
Full resolution	When you need every detail, such as reading small text or inspecting damage closely. The file is much bigger.
Up to 4096 px	Slow networks, many recipients, or older devices. Fine for an overview.
Up to 2048 px	The smallest file, for a quick overview or a very slow link.

Empty areas

A drone map is an irregular shape inside a rectangle, and the area outside the flight has no imagery.

- **Transparent** (recommended): the empty area is see-through, so the TAK basemap shows around your imagery.
- **Black**: fills the empty area with solid black. Use it only if someone you share with can't see the map properly.

Quality

A slider from **Smaller file** to **Sharper map** (10 to 95, default **70**).

- **60–75** looks good for most drone maps.
- Lower values give a smaller file, but fine detail gets soft or blocky.
- Above about 85, the file grows a lot for little visible improvement.

To judge it, use the **Detail check** on a spot with fine detail, such as text, vehicles or road lines, and compare *Original* with the compressed version beside it.

Quick picks

Situation	Image size	Quality	Empty areas
Everyday use	Up to 8192 px	70	Transparent
Slow network or many recipients	Up to 4096 px	60	Transparent
Close inspection (text, damage)	Full resolution	80	Transparent (check the size before sending)
Recipient can't see the map properly	Up to 8192 px	70	Black

Results depend on the image: busy scenes such as forest or rubble don't shrink as much as open ground or rooftops.

Sharing with your team

After converting, tap **Share...** in the Map Compressor panel. ATAK's normal send screen opens, so you can send the map to:

- **Contacts**: they get the usual "file received" message.
- **A TAK Server**, as a Data Package your team can download.
- **A Data Sync mission**, so everyone subscribed gets it.

You can also share the map later from **Overlay Manager** like any other overlay, or copy the `.tif` file from the `atak/grg` folder.

Recipient uses...	How they open it	Status
ATAK	Accept the file when it arrives, or **Import Manager** → **Image Overlay File**.	✅ Tested on 5.6 5.7 / 5.8 expected to match
iTAK / TAK Aware	Accept the file, or import it.	Please report results
WebTAK / CloudTAK	Open it from the received files or the import menu.	Please report results

Troubleshooting

Problem	What to try
Map Compressor isn't in ATAK's plugin list	Check you downloaded the file for your ATAK version (page 2). Restart ATAK and look in Settings → Tool Preferences → Plugins.
ATAK says the signature is invalid	An old test version may be installed. Uninstall Map Compressor, then install the download from page 2.
Can't find my file after tapping Choose GeoTIFF...	The file must end in `.tif` or `.tiff` and be saved on the device (for example in Download), not only in a cloud app.
The map is in the wrong place	The file may be missing its location data. Please report it, with the software that made the map.
Black box around the imagery	Set **Empty areas** to **Transparent** and convert again.
Blurry when zoomed in	Raise **Quality**, or choose a bigger **Image size**.
File still too big	Lower **Image size** first (8192 → 4096), then lower **Quality**.
Very slow, or the device runs out of memory	Very large maps (over 500 MB) need a lot of memory. Close other apps and use **Up to 8192 px** or smaller.

Questions, problems or feedback?

Reply to the email this guide came with, or report it at github.com/AM-Ripple2/map-compressor/issues. It helps to include your device, which TAK app and version you use, the map file's size, and the software that made it.